LECTURE NOTES

Topic 3: Design Solutions

Learning Outcomes:

(a) Define algorithm, pseudocode and flowchart. (1st hour)

LECTURE NOTES

Example of algorithms

- Facial recognition algorithm
- Online advertisement recommendation
  - Often if you search for a product, more product of the same category will appear
- Social media recommendation

LECTURE NOTES

Example of algorithms in daily life

Waking up going to class
Preparing an instant noodle
Preparing a cup of coffee

*Note that there are varieties of ways in achieving the same output
*It can be very simple or super complex depending on how you think

LECTURE NOTES

Definition of an algorithm

A step by step instructions that convert input to output to solve the problems.

LECTURE NOTES

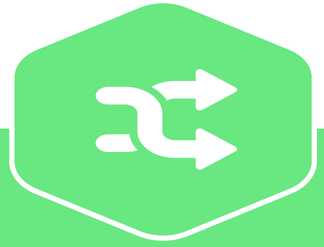


Algorithm

A sequence of well-defined steps to solve a problem.

Two **techniques** that can be used to represent algorithm:

- **Pseudocode**
- **Flowchart**



A **good algorithm** must satisfy these requirements:

- Input and output should be **defined precisely**.
- Each step in algorithm should be **clear** and **unambiguous**
- It must be **general** (can be used for different inputs)



The **purpose** of algorithm:

- To provide clear understanding of the problem.
- To provide the computer, or rather the software, a set of instructions to follow, which must be complete till the end, and in the order provided.
- To consistently determine the actions taken as a result an investigation.

LECTURE NOTES

Algorithm representation

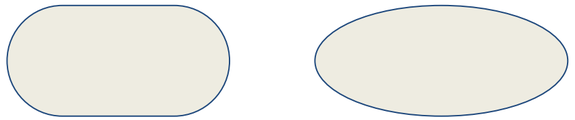
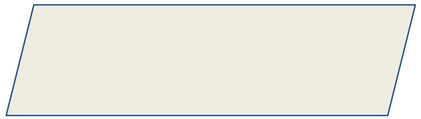
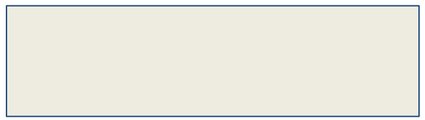
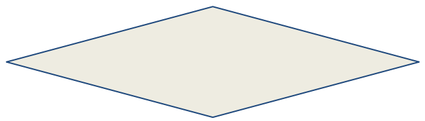
An algorithm can be represented either
- Pseudocode
- Flowchart

LECTURE NOTES

` Pseudocode	vs	Flowchart
• Informal language using English like language to design algorithms • A notation resembling a simplified programming language, used in program design		• A graphical representation of a algorithm in relation to its sequence of functions • Special-purpose symbols connected by arrows

LECTURE NOTES

Basic Flowchart component

	Capsule/Ellipse	Used to represent start-stop OR begin-end
	Arrow	Indicates the flow of the algorithm
	Parallelogram	Represents an input or output
	Rectangle	Represents a process
	Diamond	Indicates a decision

LECTURE NOTES

- **Sequence Control Structure**
  - A series of actions that are **sequentially executed** in the order they are written.

    (***example*** : Calculate sum of two numbers)

LECTURE NOTES

- **Selection Control Structure**
  - Decide which statements to execute based on a condition
  - Defines two courses of action depending on the **outcome** of a **condition**. (***example*** : Determine "Pass" or "Fail" based on grade entered)

LECTURE NOTES

- **Repetition/ Looping / Iteration Control Structure**
  - used to tell a program to execute statements repeatedly
  - Specifies a block of one or more statements that are **repeatedly** executed until a condition is satisfied.

    (***example*** : Display "Hello" for 10 times)

LECTURE NOTES

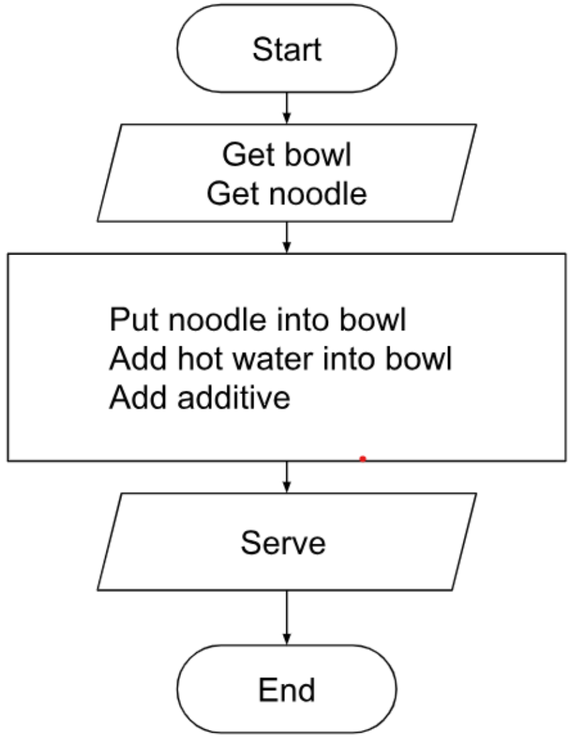
Example 1: Algorithm example in daily life

Problem statement: Preparing an instant noodle

Pseudocode

```
Start
    Get bowl
    Get noodle
    Put noodle into bowl
    Add hot water into bowl
    Add additive
    Serve
End
```

Flowchart



LECTURE NOTES

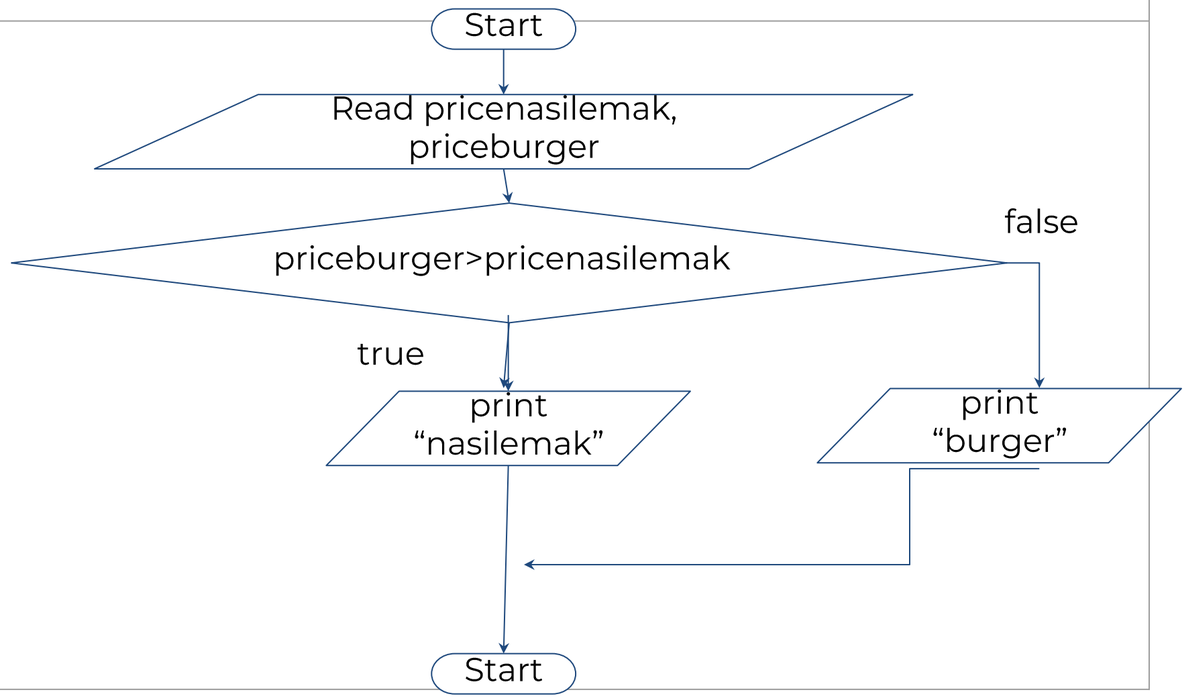
Example 2: Algorithm example in daily life

Problem statement: If burger is more expensive than nasi lemak, choose nasi lemak (selection)

Pseudocode

```
Start
    Get pricenasilemak
    Get priceburger
    if priceburger > pricenasilemak
      print “nasilemak’
    else
      print “burger”
    end if
End
```

Flowchart

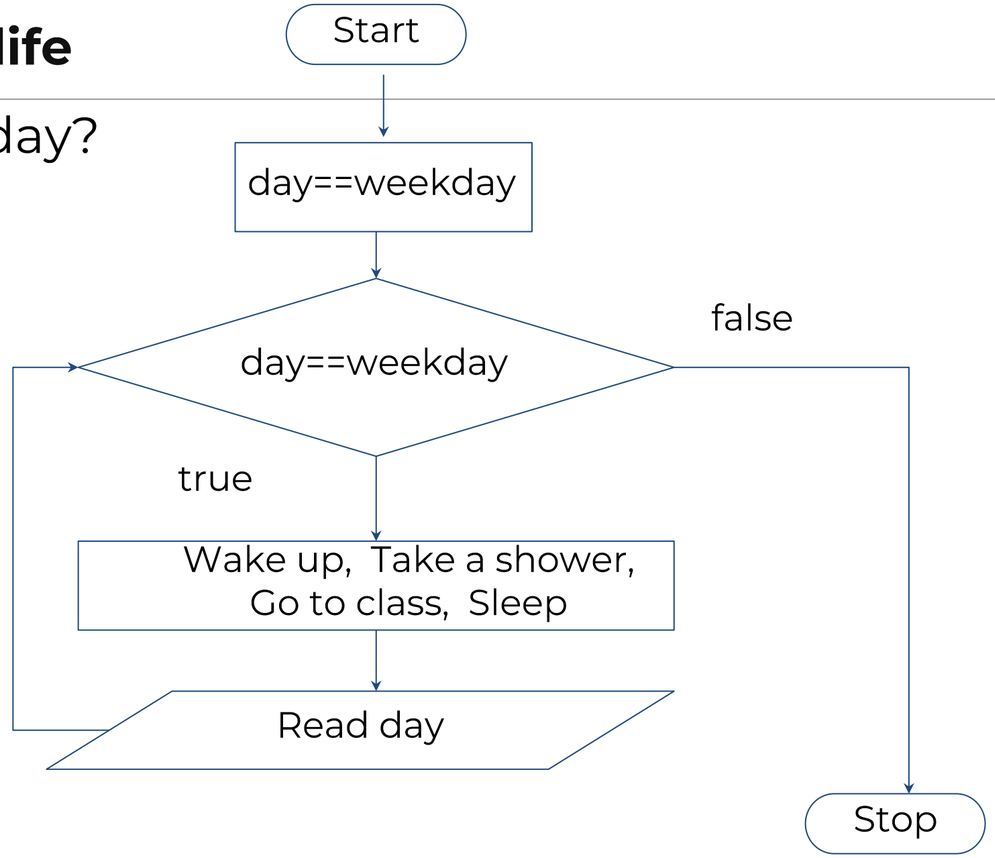


LECTURE NOTES

Example 3: Algorithm example in daily life

Problem statement: What do you do everyday?

```
Start
    while weekday
      Wake up
      Take a shower
      Go to class
      Sleep
      Repeat until end of weekday
End
```



TUTORIAL NOTES

Topic 3: Design Solutions

3.1.1 : Sequence Control Structure

Learning Outcomes:

a. Describe sequence control structure using algorithm.
b. Solve a given problem using pseudocode for sequence structure.
c. Solve a given problem using flowchart for sequence structure.

TUTORIAL NOTES

Sequence Control Structure

A series of actions that are **sequentially executed** in the order they are written.
(***example*** : Calculate sum of two numbers)

TUTORIAL NOTES

Problem statement 1

Find the sum of three numbers.

IPO model

TUTORIAL NOTES

Problem statement 1

Find the sum of three numbers.

IPO model

Input: number1, number2, number3
Process: Calculate the sum based on number1, number2 and number3.
Output : sum

Pseudocode

TUTORIAL NOTES

Problem statement 1

Find the sum of three numbers.

IPO model

Input: number1, number2, number3
Process: Calculate the sum based on number1, number2 and number3.
Output : sum

Pseudocode

```
Start
    Read number1, number2, number3
    sum = number1 + number2 + number3
    Print sum
Stop
```

TUTORIAL NOTES

Problem statement 1

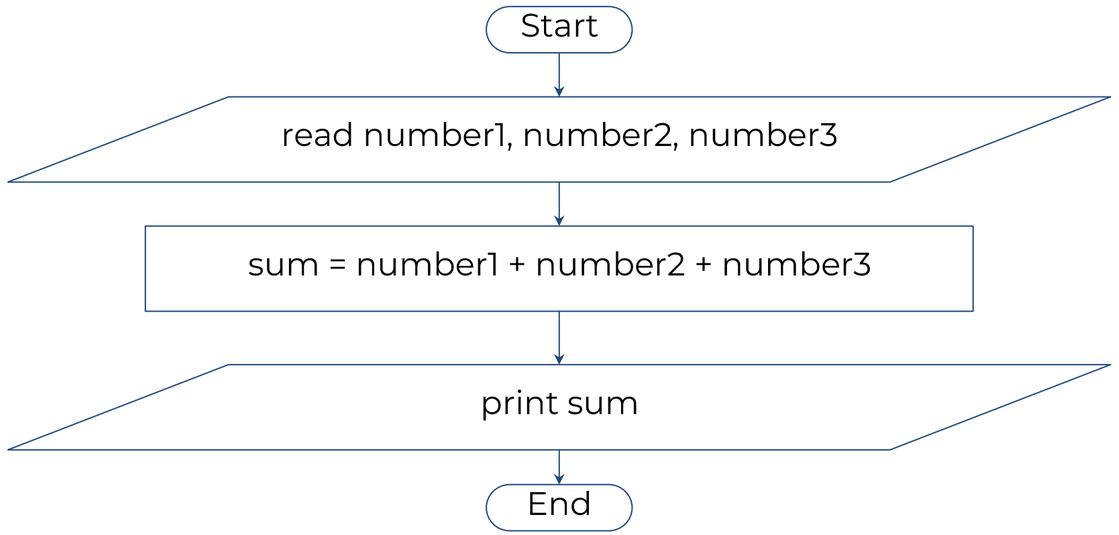
Find the sum of three numbers.

Pseudocode	Flowchart
Start read number1, number2, number3 sum = number1 + number2 + number3 print sum Stop	

TUTORIAL NOTES

Problem statement 1

Find the sum of three numbers.

Pseudocode	Flowchart
Start read number1, number2, number3 sum = number1 + number2 + number3 print sum Stop	

TUTORIAL NOTES

Problem statement 2

Find the average of three numbers.

TUTORIAL NOTES

Problem statement 2 (Sequence control structure)
Find the average of three numbers.

IPO model
Input: number1, number2, number3
Process: Calculate the average based on number1, number2 and number3.
Output : average

TUTORIAL NOTES

Problem statement 2 (Sequence control structure)
Find the average of three numbers.

IPO model
Input: number1, number2, number3
Process: Calculate the average based on number1, number2 and number3.
Output : average

Pseudocode

```
Start
        read number1, number2, number3
        average = (number1 + number2 + number3)/3
        print average
Stop
```

TUTORIAL NOTES

Problem statement 2

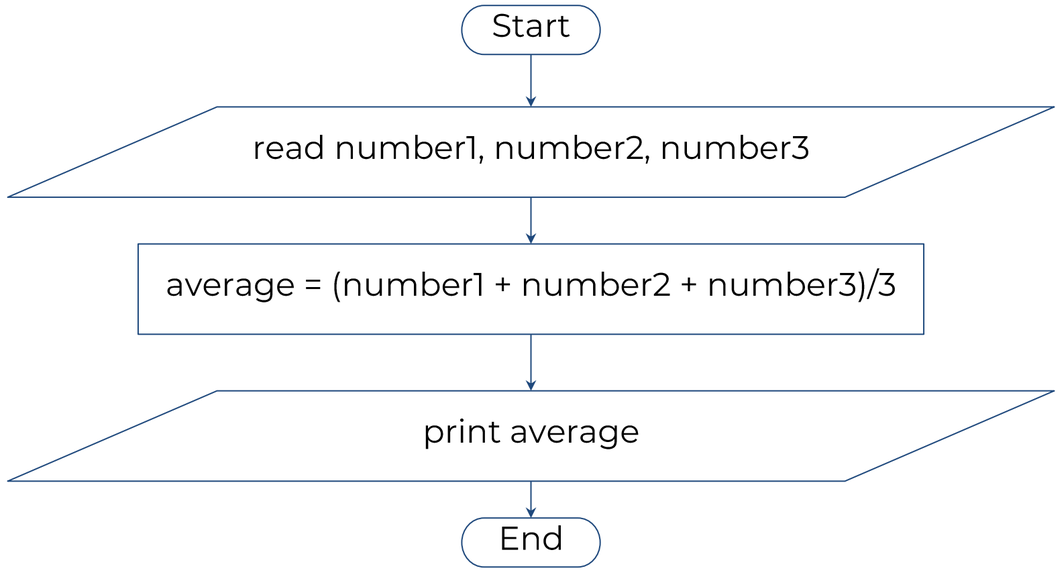
Find the average of three numbers.

Pseudocode	Flowchart
Start read number1, number2, number3 average = (number1 + number2 + number3)/3 print average Stop	

TUTORIAL NOTES

Problem statement 2

Find the average of three numbers.

Pseudocode	Flowchart
Start read number1, number2, number3 average = (number1 + number2 + number3)/3 print average Stop	

TUTORIAL NOTES

Problem Statement 3

Identify the input, process and output for the given problem statement.

The program will calculate mean of 5 tests mark

Input **:** test1, test2, test3, test4, test5

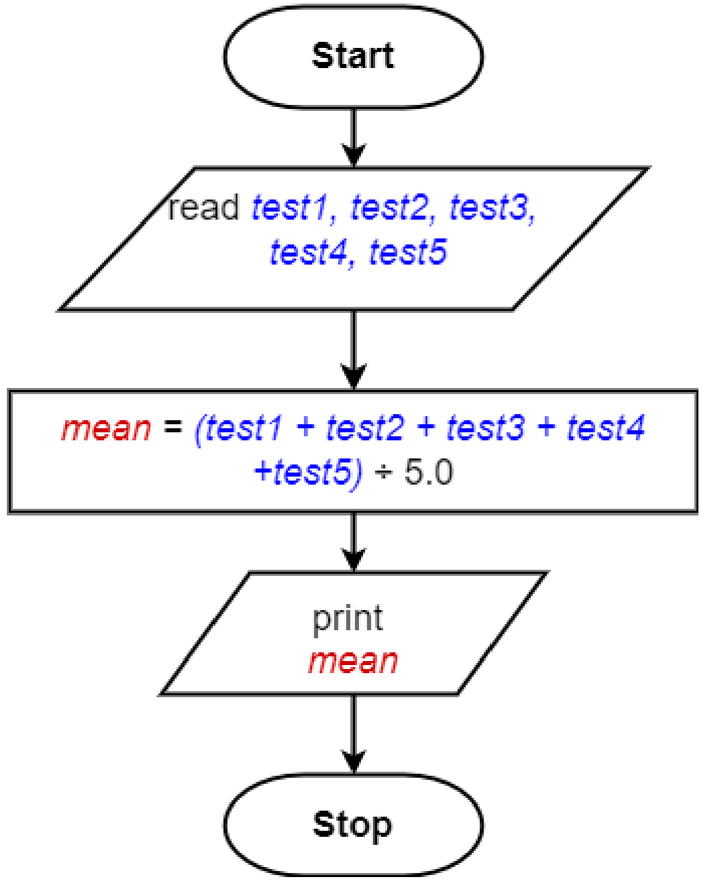
Process **:**

1. Calculate total based on test1, test2, test3, test4, test5
2. Calculate mean based on test1, test2, test3, test4, test5

Output **:** mean

TUTORIAL NOTES

Problem Statement 3

Pseudocode	Flowchart
Start read test1, test2, test3, test4, test5 mean = (test1 + test2 + test3 + test4 +test5) ÷ 5.0 print mean Stop	

TUTORIAL NOTES

Problem Statement 4

The program will find the square of a given number.

Input : *number* ✔

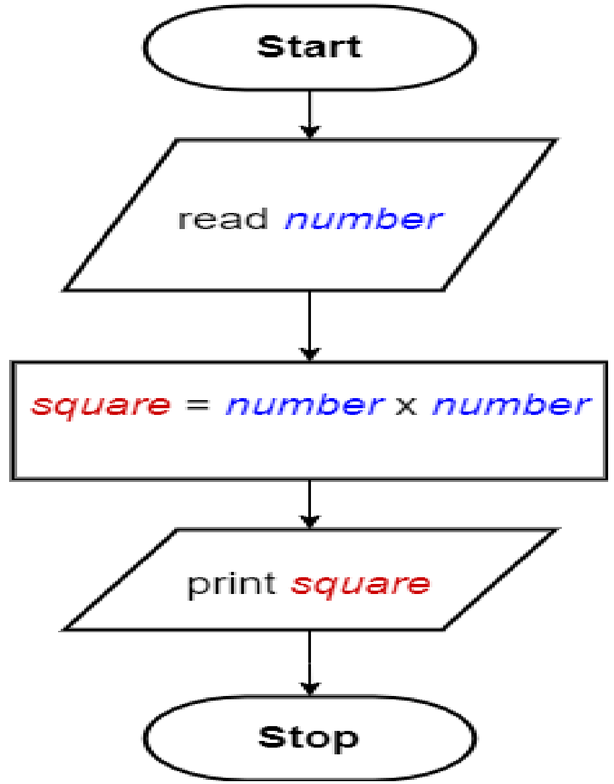
Process : Calculate *square* based on *number*

Output : *square*

TUTORIAL NOTES

Problem Statement 4

The program will find the square of a given number.

Pseudocode	Flowchart
Start read *number* *square* = *number* x *number* print *square* **Stop** **the formula must be a written in full, without superscript y*	

TUTORIAL NOTES

Topic 3: Design Solutions

3.1.2 : Selection Control Structure

Learning Outcomes:

a. Describe selection control structure using algorithm.
b. Solve a given problem using pseudocode for selection structure.
c. Solve a given problem using flowchart for selection structure.

TUTORIAL NOTES

Selection Control Structure

- Decide which statements to execute based on a condition
- Defines two courses of action depending on the **outcome** of a **condition**.

  (***example*** : Determine "Pass" or "Fail" based on grade entered)

TUTORIAL NOTES

Types of Selection Control Structure

- Single selection
- Dual selection
- Multiple selection

TUTORIAL NOTES

Single selection

Problem statement 1

Display a message **“Congratulations! First Class Students”** for a student whose cumulative grade point average (CGPA) is 3.70 or above.

TUTORIAL NOTES

Single selection

Problem statement 1

Display a message **“Congratulations! First Class Students”** for a student whose cumulative grade point average (CGPA) is 3.70 or above.

IPO model

Input: CGPA
Process: To determine message based on CGPA.
Output: “Congratulations! First Class Students” or no output

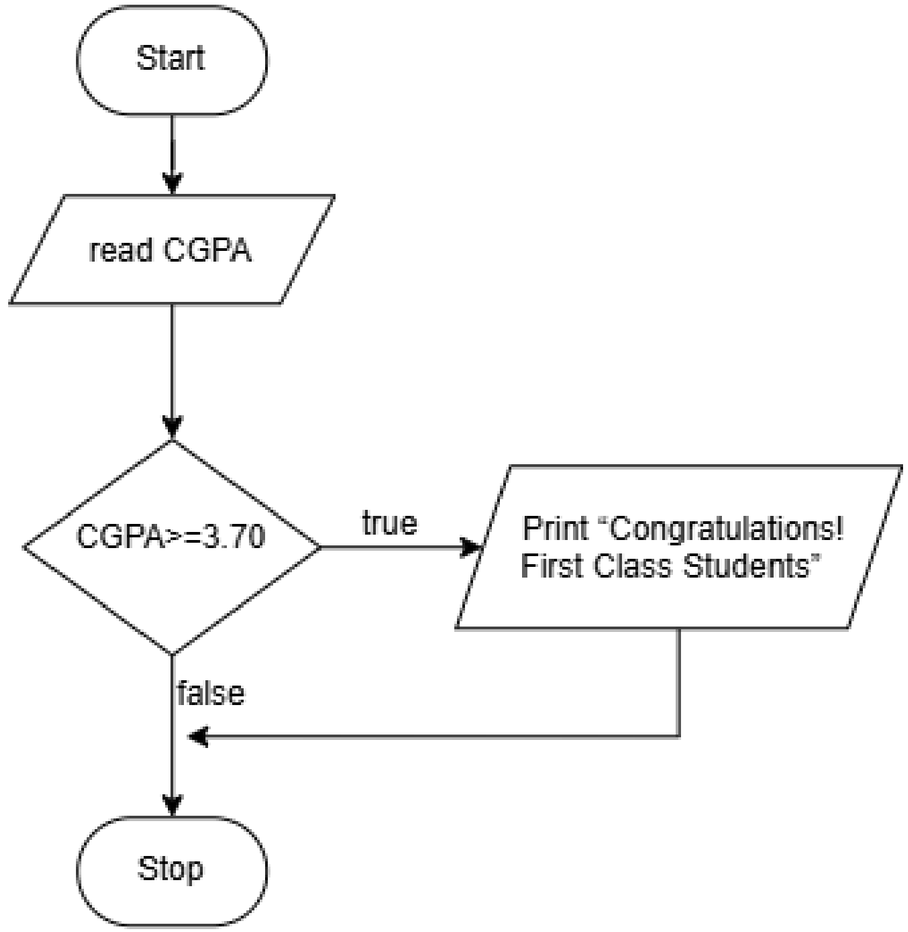
TUTORIAL NOTES

Single selection

Pseudocode

```
Start
    read CGPA
    if CGPA >=3.70
          print “Congratulations! First Class Students”
    end if
Stop
```

Flowchart



TUTORIAL NOTES

Dual selection

Problem statement 2

Determine the smaller number between two numbers.

TUTORIAL NOTES

Dual selection

Problem statement 2

Determine the smaller number between two numbers.

IPO model

Input: number1, number2
Process: Determine the smaller number based on number1 and number2.
Output : smaller

TUTORIAL NOTES

Dual selection

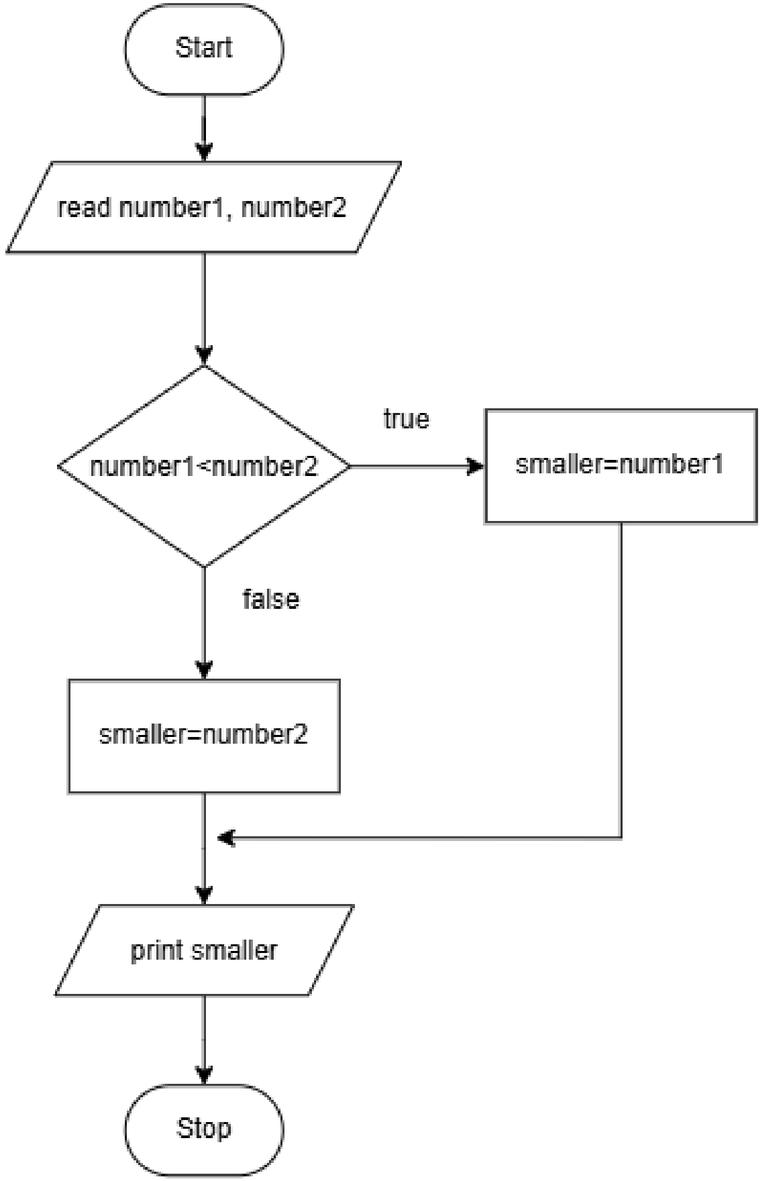
Problem statement 2

Determine the smaller number between two numbers.

Pseudocode

```
Start
        read number1, number2
        if number1 < number2
                smaller = number1
        else
                smaller = number2
       end if
        print smaller
Stop
```

Flowchart



TUTORIAL NOTES

Multiple selection

Problem statement 3

Create a program that will ask a user to enter age for two people. Display “The first person is older” when age1 is greater than age2, else display “The second person is older”. If both are not true, display “Both persons are same age”.

TUTORIAL NOTES

Multiple selection

Problem statement 3

Create a program that will ask a user to enter age for two people. Display “The first person is older” when age1 is greater than age2, else display “The second person is older”. If both are not true, display “Both persons are same age”.

IPO model

Input : age1, age2
Process : Determine respective message based on age1 and age2.
Output : “The first person is older” or “The second person is older” or “Both persons are same age”

TUTORIAL NOTES

Multiple selection

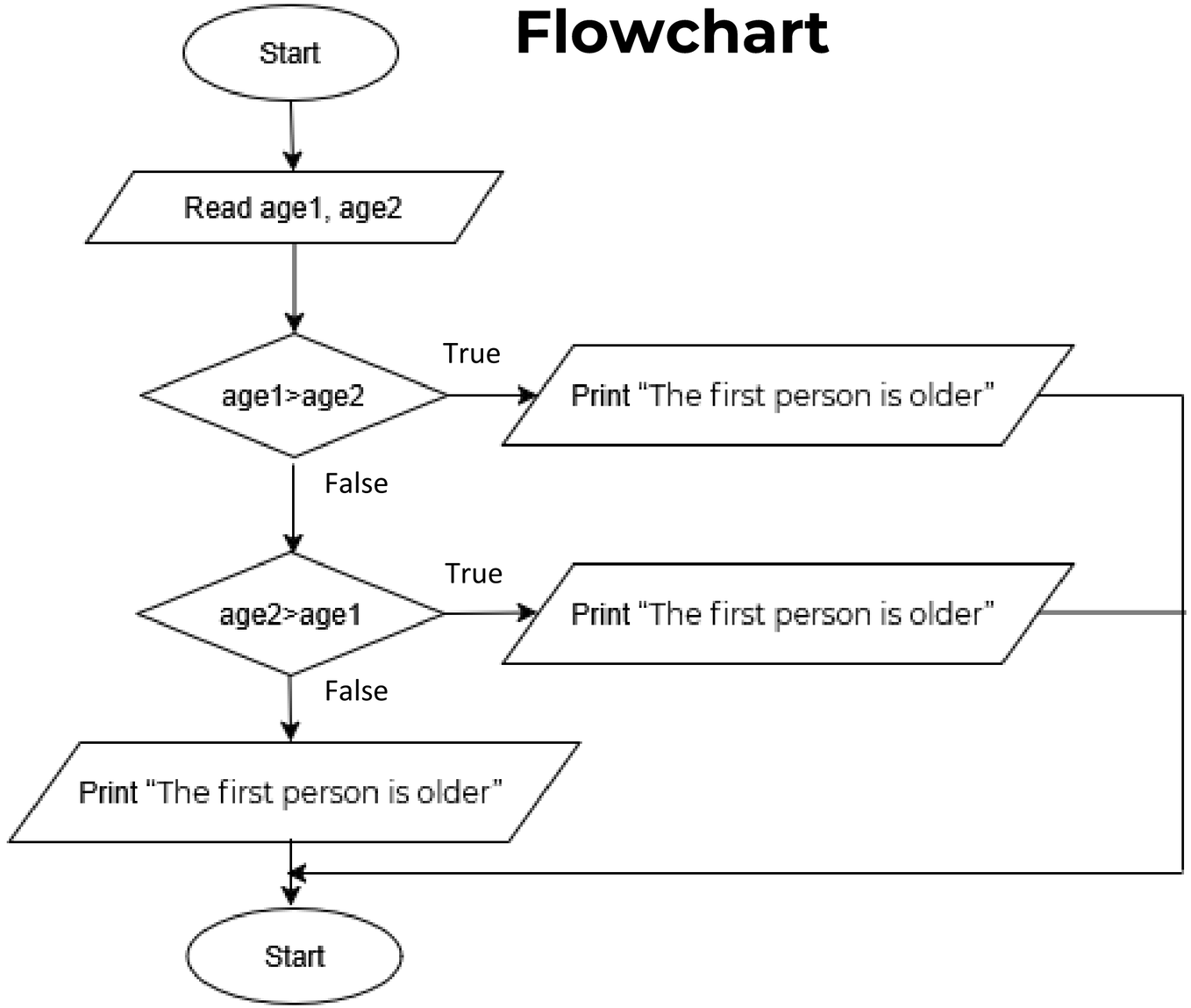
Pseudocode

```
Start
        read age1, age2
        if age1>age2
                print “The first person is older”
        else if age2>age1
                print “The second person is older”
        else
                print “Both persons are same age”
        endif
Stop
```

TUTORIAL NOTES

Multiple selection

Flowchart



TUTORIAL NOTES

Topic 3: Design Solutions

3.1.2 : Selection Control Structure

Learning Outcomes:

a. Describe repetition control structure using algorithm.
b. Solve a given problem using pseudocode for repetition structure.
c. Solve a given problem using flowchart for repetition structure.

TUTORIAL NOTES

Repetition Control Structure

- used to tell a program to execute statements repeatedly
- Specifies a block of one or more statements that are **repeatedly** executed until a condition is satisfied.
  (***example*** : Display "Hello" for 10 times)

TUTORIAL NOTES

Types of Repetition Control Structure

- Counter-controlled
- Sentinel-controlled

TUTORIAL NOTES

CS125

Repetition: Counter controlled

In **Counter-Controlled Repetition**, the number of loops that need to be executed is known (either specified in the question or determined by the user), and it depends on the value specified in the condition.

IPO Analysis General Format		*Pseudocode General Format*
Input	**input for ____ times**	**Start** counter initialization **while (condition)** statement block counter increment/decrement **end while** next action (after the loop has ended) **Stop** ***The statement(s) will be executed repeatedly until the condition is False (F)*** **Use correct **indentation**. Avoid writing actions from the same starting line of Start and Stop.* *Note: end while is used to indicate the end of an if block. The condition (Boolean expression) must be written in bracket( ).*
Process	**Repeat calculate *output* based on input for ____ times**	
Output	***output* for ____ times**	

TUTORIAL NOTES

Repetition: Counter controlled (Other approach for IPO)

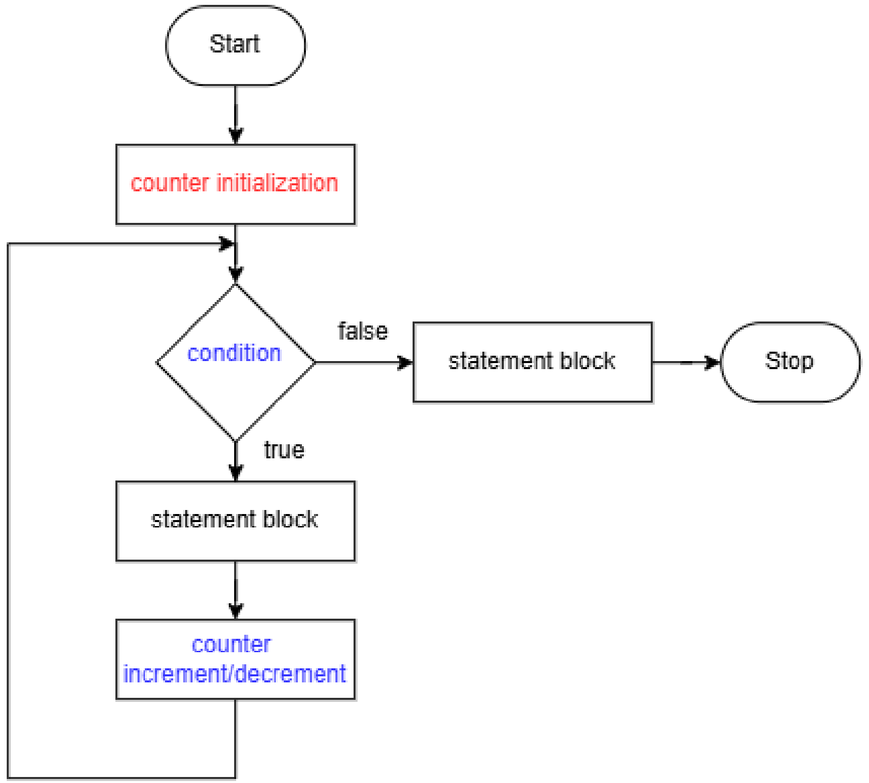
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Process	**Repeat calculate *output* based on input for ____ times**	
Output	***output***	

TUTORIAL NOTES

Repetition: Counter controlled - Counter controlled

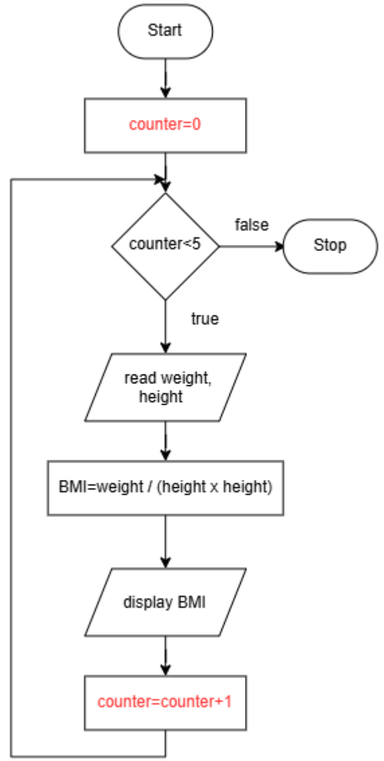
Flowchart General Format



TUTORIAL NOTES

Problem statement 1- Counter controlled

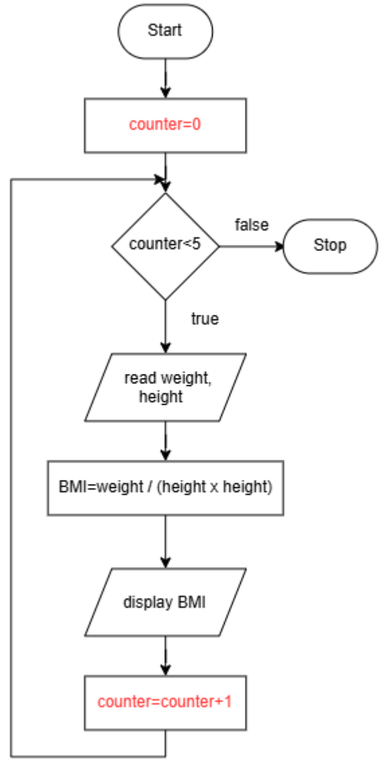
The program calculate BMI for 5 persons

IPO Analysis	Pseudocode	Flowchart
Input : *weight*, *height* **for 5 times** Process : **Repeat calculate BMI based on** *weight* and *height* **for 5 times** Output : **BMI for 5 times**	**Start** counter = 0 while (counter < 5) read weight, height BMI = weight ÷ (height x height) display BMI counter = counter + 1 end while **Stop** **Note**: *To ensure the loops are repeated for 5 times:* *-if the counter starts with 0, the condition is* while (counter < 5) *-if the counter starts with 1, the condition is* while (counter ≤ 5) @ while (counter < 6)	

TUTORIAL NOTES

Problem statement 1- Counter controlled (Other approach for IPO)

The program calculate BMI for 5 persons

IPO Analysis	Pseudocode	Flowchart
Input : *weight*, *height* Process : repeat input weight, height for 5 times **Repeat calculate BMI based on** *weight* and *height* **for 5 times** Output : **BMI** **standard penulisan : penggunaan input dalam proses dengan pernyataan "based on (input)"**	**Start** counter = 0 while (counter < 5) read weight, height BMI = weight ÷ (height x height) display BMI counter = counter + 1 end while **Stop** **Note**: *To ensure the loops are repeated for 5 times:* *-if the counter starts with 0, the condition is* while (counter < 5) *-if the counter starts with 1, the condition is* while (counter ≤ 5) @ while (counter < 6)	

TUTORIAL NOTES

Problem statement 2 - Counter controlled

Calculate the **average** BMI of five students.

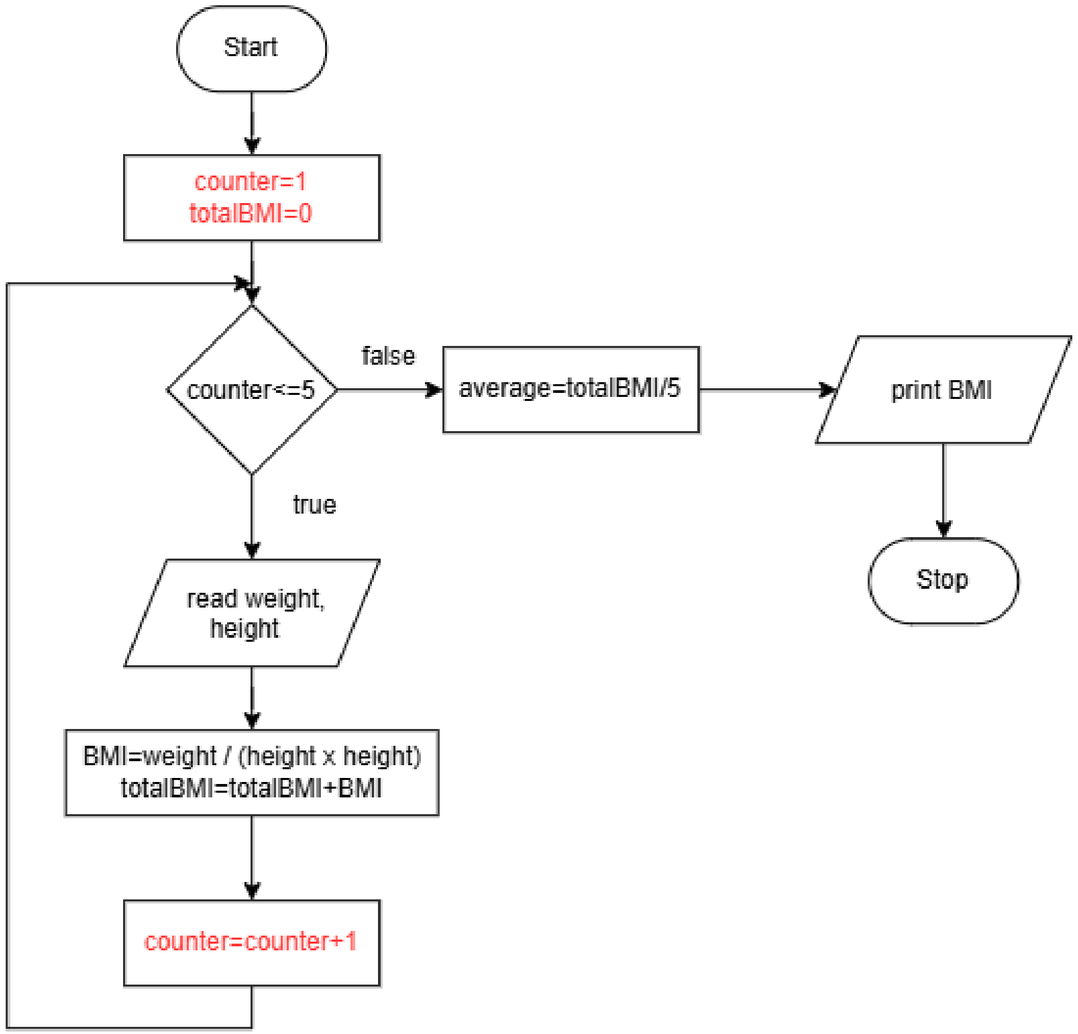
IPO model

Input : weight, height for five students
Process: Repeat to calculate total BMI for five students based on weight and height. Calculate the average BMI based on total BMI.
Output : average BMI

TUTORIAL NOTES

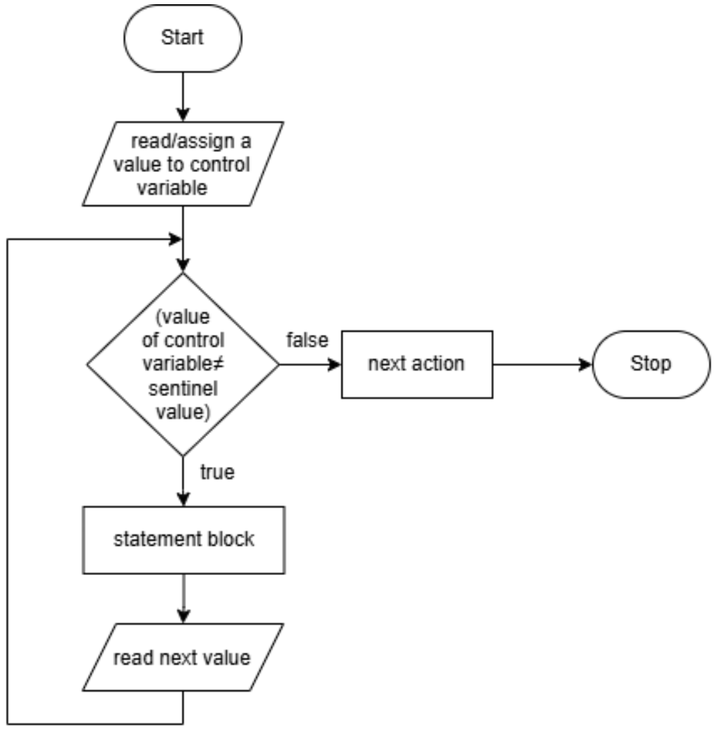
Flowchart

```
Start
    counter = 1
    totalBMI = 0
    while (counter <= 5)
        read weight, height
        BMI = weight / (height x height)
        totalBMI = totalBMI + BMI
        counter = counter + 1
    end while
    averageBMI = totalBMI/5
    print averageBMI
Stop
```



TUTORIAL NOTES

Repetition: Sentinel controlled

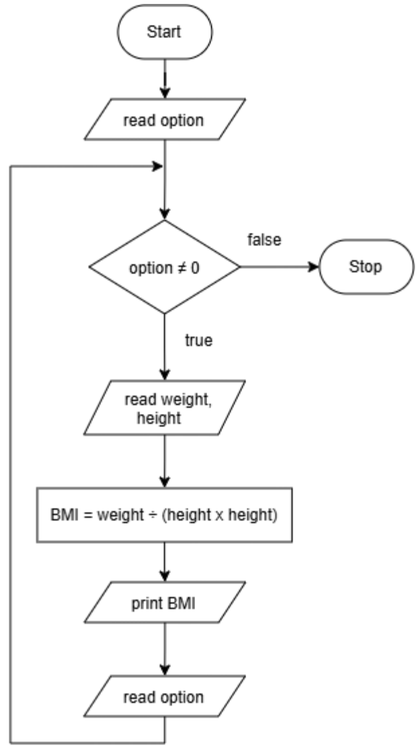
Pseudocode General Format	*Flowchart General Format*
Start read / Assign a value to control variable while (value of control variable is not sentinel value) statement block read next value end while next action Stop **Note:The statement(s) will be executed repeatedly until the condition is False (F)**	

TUTORIAL NOTES

CS125

Problem statement 2 - Sentinel controlled

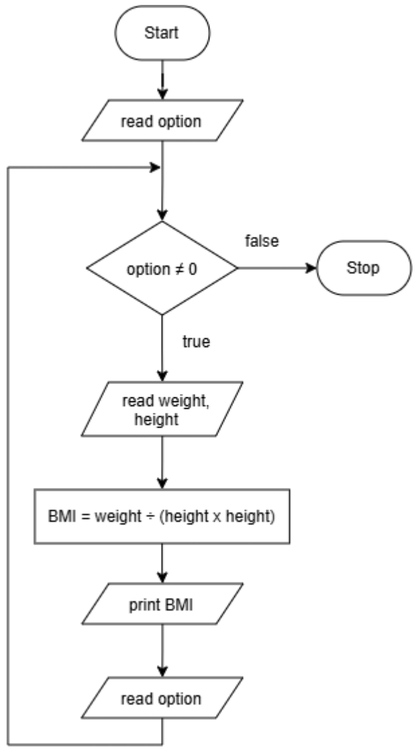
The program calculates BMI of persons and stops when the user enters 0.

IPO Analysis	Pseudocode	Flowchart
Input : ***option*, *weight, height unit user enter 0*** Process : **Repeat** calculate **BMI for *x times unit user enter 0*** Output : **BMI for *x times unit user enter 0***	**Start** read option while (option ≠ 0) read weight, height BMI = weight ÷ (height x height) display BMI read option end while **Stop**	

TUTORIAL NOTES

Problem statement 2 - Sentinel controlled (Other approach for IPO)

The program calculates BMI of persons and stops when the user enters 0.

IPO Analysis	Pseudocode	Flowchart
Input : ***option*, *weight, height*** Process : **Repeat** calculate **BMI** **based on weight and height until option equal to 0 (remove for statement)** Output : **BMI** **PENULISAN CONDITION** **statement while (option not equal 0) dalam bentuk ayat juga boleh diterima dalam pseudocode	**Start** read option while (option ≠ 0) read weight, height BMI = weight ÷ (height x height) display BMI read option end while **Stop (indentation)**	

** statement coding !=, ==, “A” boleh diterima

TUTORIAL NOTES

Repetition - Sentinel controlled

Calculate the total of several numbers entered by the user and end the calculation when the user enters 0.

IPO model

Input: number until number entered is 0
Process: Repeat to calculate total of several numbers based number until number entered is 0.
Output : total

TUTORIAL NOTES

Pseudocode

```
Start
    total = 0
    read number
    while (number != 0)
        total = total + number
        read number
        end while
    print total
Stop
```

Flowchart

